

Mathematics, 2013 EC, natural science, entrance exam, round 1

Grade 12 · Mathematics · 2013 E.C.

Time allowed: 120 minutes

Questions: 55

Total marks: 55

Name _____ Grade _____ ID No. _____ Date _____

1. Given two vectors $u = (1, 3)$ and $v = (-3, 5)$ in the plane. Then $6u + 2v$ is equal to (1)
 - A. (0, 18)
 - B. (3, 3)
 - C. (2, 28)
 - D. (0, 28)
2. Which one of the following is the standard form of the equation of the circle centered at $(2, -3)$ and radius 5? (1)
 - A. $(x - 2)^2 + (y - 3)^2 = 25$
 - B. $(x - 2)^2 + (y + 3)^2 = 25$
 - C. $(x + 2)^2 + (y + 3)^2 = 5$
 - D. $(x + 2)^2 + (y - 3)^2 = 25$
3. If f is a continuous function on $[a, b]$ and F is its anti-derivative, then which of the following is true? (1)
 - A. $\int_a^b F'(x)dx = f(b) - f(a)$
 - B. $\int_a^b F(x)dx = f(b) - f(a)$
 - C. $\int_a^b f(x)dx = F(b) - F(a)$
 - D. $\int_a^b f'(x)dx = F(b) - F(a)$
4. Which of the following defines the equation of a sphere whose center is at $(0, -1, 2)$ and radius 6 units? (1)
 - A. $x^2 + (y + 1)^2 + (z - 2)^2 = 36$
 - B. $x^2 + (y - 1)^2 + (z - 2)^2 = 36$
 - C. $x^2 + (y + 1)^2 + (z - 2)^2 = 12$
 - D. $x^2 + (y + 1)^2 + (z + 2)^2 = 36$
5. Which of the following represents the vector $(3, -5, 7)$ in the space using the standard unit vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$? (1)
 - A. $3\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}$
 - B. $3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$
 - C. $3\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$
 - D. $-3\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}$
6. Which one of the following is the scalar (dot) product of the vectors $(2, -4, 6)$ and $(-2, -1, 0)$? (1)
 - A. 0
 - B. -8
 - C. 8
 - D. -4
7. What are the domain and range of the function $f(x) = 2x^{\frac{3}{4}}$ respectively? (1)
 - A. $\mathbb{R}$ and $[0, \infty)$
 - B. $[0, \infty)$ and $\mathbb{R}$
 - C. $[0, \infty)$ and $\mathbb{R}$
 - D. $[0, \infty)$ and $[0, \infty)$
8. Which one of the following defines a one-to-one function? (1)
 - A. $f = \{(x, y) : x \text{ is a sister of } y\}$
 - B. $f = \{(x, y) : y = 3x - 1\}$
 - C. $f = \{(x, y) : y = x^2 + 1\}$
 - D. $f = \{(x, y) : y \text{ is a father of } x\}$
9. The equation of a circle is given by $(x - 3)^2 + (y + 2)^2 = 9$. What are the center and the radius of the circle respectively? (1)
 - A. (3, -2) and 3
 - B. (3, 2) and 3
 - C. (3, -2) and 9
 - D. (3, 2) and 9

20. Consider the system of linear equation $\begin{cases} 3x - ay = 1 \\ 6x + 4y = 5 \end{cases}$. Then which of the following is true? (1)
- A. If $a > -2$, then the system has a unique solution.
 B. If $a = -2$, then the system has a unique solution.
 C. If $a = 2$, then the system has no solution.
 D. If $a > 2$, then the system has infinitely many solutions.
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21. What is the magnitude of the vector whose initial point is at (1, -3) and terminal point is at (4, 1)? (1)
- A. 5
 B. 6
 C. 1
 D. 4
-
22. Which of the following is true? (1)
- A. The domain of $y = \tan^{-1} x$ is $(-\infty, \infty)$.
 B. The domain of $y = \cos^{-1} x$ is $(-1, 1)$.
 C. $\sin^{-1}(0) = \frac{\pi}{2}$.
 D. $\tan^{-1}(0) = 1$.
-
23. Which one of the following is the fourth term of the sequence $\left\{ \cos\left(\frac{n\pi}{2}\right) \right\}$? (1)
- A. 0
 B. -1
 C. 1
 D. 0.5
-
24. A factory that produces electric cable had sales Birr 1,000,000 the first day. During each successive days the sales increased by Birr 10,000 every day. What is the total sales of the factory in the first 10 days? (1)
- A. Birr 10,000,000
 B. Birr 1,090,000
 C. Birr 20,900,000
 D. Birr 10,450,000
-
25. What is the least upper bound of the sequence $\left\{ 2 + \frac{n}{n+1} \right\}_{n=1}^{\infty}$? (1)
- A. 1
 B. 0
 C. 3
 D. 2
-
26. Suppose a_n and b_n are sequences with $\lim a_n = L$ and $\lim b_n = M$. Then which of the following is NOT necessarily true? (1)
- A. $\lim_{n \rightarrow \infty} (3a_n + 2b_n) = 3L + 2M$
 B. $\lim_{n \rightarrow \infty} (a_n - b_n) = L - M$
 C. $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{L}{M}$
 D. $\lim_{n \rightarrow \infty} a_n b_n = LM$
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27. Which of the following is the same as the derivative of a function f' at a point P , where P is on the graph of f ? (1)
- A. The continuity of the function at P .
 B. The slope of a secant line through P .
 C. The slope of the tangent line through P .
 D. The graph of f has no jump at P .
-
28. What is the derivative of the function $f(x) = 3x^2 + 2\sqrt{x} - 4x$ at $x = 1$? (1)
- A. -1
 B. -3
 C. 3
 D. 1

29. Which of the following is true about the zero of a function? (1)
- A. It is the y-intercept of the function.
 - B. It is the x-intercept of the function.
 - C. It is the concavity point of the function.
 - D. It is the critical point of the function.
-
30. Which of the following is true about the function defined by $f(x) = x^3 - 3x^2$ on $[-1, 3]$? (1)
- A. Its critical numbers are 0 and 2 and its maximum value is 15.
 - B. Its critical numbers are 0, 2, 3 and its maximum value is 0.
 - C. It has critical numbers at 0 and 2 and its maximum value is 0.
 - D. Its critical numbers are 0 and 2 and its maximum value is -4.
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31. What is the number that satisfies Rolle's Theorem for $f(x) = x^3 - 3x + 1$ on $[0, 3]$? (1)
- A. 1
 - B. $-\frac{3}{2}$
 - C. 2
 - D. $\frac{3}{2}$
-
32. What are the length L and width W of a rectangle with perimeter 10,000m that maximizes the area? (1)
- A. L = 2,500 m and W = 2,000 m
 - B. L = 2,500 m and W = 2,500 m
 - C. L = 5,000 m and W = 5,000 m.
 - D. L = 3,000 m and W = 2,500 m.
-
33. If the radius r of a sphere is increasing at the rate of 2 cm/min, then the rate of change of the volume when r = 1 cm is (1)
- A. $6\pi \text{ cm}^3/\text{min}$
 - B. $4\pi \text{ cm}^3/\text{min}$
 - C. $12\pi \text{ cm}^3/\text{min}$
 - D. $8\pi \text{ cm}^3/\text{min}$
-
34. Suppose $\int_1^3 f(x) dx = 6$ and $\int_1^2 f(x) dx = 10$, what is the value of $\int_2^3 f(x) dx$? (1)
- A. 2
 - B. -4
 - C. 16
 - D. 4
-
35. Which of the following statements has a truth value T? (1)
- A. $(p \wedge q) \Rightarrow (p \vee q)$
 - B. $(p \wedge q) \Rightarrow \neg(p \vee q)$
 - C. $(p \vee q) \Rightarrow (p \wedge q)$
 - D. $(p \Rightarrow q) \Rightarrow (p \vee q)$
-
36. Which of the following is equal to $\int (\cos x - 2x) dx$ where c is a constant? (1)
- A. $\sin x + x^2 + c$
 - B. $-\sin x - x^2 + c$
 - C. $\sin x - x^2 + c$
 - D. $-\sin x + x^2 + c$
-
37. What is the area of the region bounded by $y = 0$ and the graph of $f(x) = 4 - x^2$ on the interval $[-1, 1]$? (1)
- A. $\frac{22}{3}$
 - B. $\frac{2}{3}$
 - C. $\frac{26}{3}$
 - D. 8

38. The inverse of the function $y = 3x - 5$ is equal to (1)

A. $y = \frac{x+5}{3}$
C. $y = \frac{x-5}{3}$

B. $y = \frac{x+3}{5}$
D. $y = -5x + 3$

39. Which of the following is the universal set of the expression $\frac{1}{x^2} \div \frac{x+1}{x^2+x}$? (1)

A. $\mathbb{R} \setminus \{0, 1\}$
C. $\mathbb{R} \setminus \{0, 1, -1\}$

B. $\mathbb{R} \setminus \{0\}$
D. $\mathbb{R} \setminus \{0, -1\}$

40. The equation of a line is given by $3x - 2y + 6 = 0$. Then which of the following is true about the line? (1)

A. Its y-intercept is -3
C. $(2, 5)$ lies on the line

B. Its slope is $\frac{3}{2}$
D. Its x-intercept is 2

41. Which one of the following is the standard form of the equation of the parabola with vertex $(4, 6)$ and focus $(-8, 6)$? (1)

A. $(x - 4)^2 = -12(y - 6)$
C. $(y - 6)^2 = 48(x - 4)$

B. $(x - 4)^2 = 12(y - 6)$
D. $(y - 6)^2 = -48(x - 4)$

42. Which of the following is true about the hyperbola given by $\frac{x^2}{4} - y^2 = 1$? (1)

A. Its foci are at $(-5, 0)$ and $(5, 0)$
C. Its center is at $(2, 1)$

B. Its vertices are at $(-2, 0)$ and $(2, 0)$
D. An asymptote to the hyperbola is $y = 2x$

43. Two statements p and q are given by: $p : \frac{22}{7}$ is an irrational number. q : The sum of two odd integers is odd. Then which of the following has a truth value T ? (1)

A. $p \wedge \neg q$
C. $\neg p \wedge q$

B. $p \vee \neg q$
D. $p \wedge q$

44. Which one of the following is the contrapositive of the statement: If 21 is a prime number, then 4 is an odd number? (1)

A. If 4 is an odd number, then 21 is not a prime number
B. If 4 is an odd number, then 21 is a prime number
C. If 4 is not an odd number, then 21 is not a prime number
D. If 21 is not a prime number, then 4 is an even number

45. Which of the following is true where c is a constant? (1)

A. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = -2 \cos \sqrt{x} + c$

B. $\int (3x + 5)^6 dx = \frac{(3x + 5)^7}{7} + c$

C. $\int \ln x dx = \frac{1}{x} + c$

D. $\int x e^x dx = \frac{x^2}{2} e^x + c$

46. The acceleration of a body at any time t is given by $a(t) = 4 - 2t + 3t^2$ with initial velocity $v(0) = 5$. Then what is the velocity v of the body as a function of time? (1)

A. $v(t) = 1 + 4t - t^2 + t^3$
C. $v(t) = -2 + 6t$

B. $v(t) = 5 + 4t - t^2 + t^3$
D. $v(t) = 9 - t^2 + t^3$

47. Which of the following argument is valid? (1)
- A. $p, \neg q \vdash p \implies q$ B. $\neg p, q \vdash p \implies q$
 C. $p, \neg q \vdash p \implies q$ D. $p, \neg q \vdash p \wedge q$
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48. Which of the following proof is more suitable to prove the assertion: $1 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2$ (1)
- A. Proof by counter example
 B. Proof by the principle of mathematical induction
 C. Proof by contradiction
 D. Proof by indirect proof
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49. What is the coefficient of x^2 in the binomial expansion of $(2x + \frac{3}{x^3})^8$? (1)
- A. 6 B. 0
 C. 576 D. 500
-
50. Which of the following is true about the graph of $y = -4 \cos(\frac{2x}{3})$? (1)
- A. Its period is 3π
 B. Its amplitude is -4
 C. The graph completes one cycle on the interval $[0, \frac{3\pi}{2}]$
 D. The graph completes one cycle on the interval $[0, \frac{2\pi}{3}]$
-
51. Math test scores distribution of 20 students is given as in Table 2 below? Scores 1 - 5 6 - 10 11 - 15 16 - 20 (1)
 Frequency 4 6 8 2 What is the standard deviation of the scores?
- A. 5 B. 4
 C. $\sqrt{21}$ D. $\sqrt{20}$
-
52. The angle of elevation of the top of a tree is found to be 60° as measured from a point on a level ground. What is the height of the tree if the angle of elevation of a point on the tree that is 8m below the top is 30° as measured from the same point on the ground? (1)
- A. 8m B. 4m
 C. 14m D. 12m
-
53. Which of the following series is convergent? (1)
- A. $\sum_{n=1}^{\infty} 5^{11-2n}$ B. $\sum_{n=1}^{\infty} (\frac{2}{5})^{-n}$
 C. $\sum_{n=1}^{\infty} (\frac{3}{2})^n$ D. $\sum_{n=1}^{\infty} 2^{2n-5}$
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54. Let f be a function defined on a closed bounded interval $[a, b]$. Which of the following describes that f is continuous on $[a, b]$? (1)
- A. f is continuous on (a, b) , f is continuous from the right at a and from the left at b
 B. f is continuous at each point in (a, b) and at both a and b
 C. f is continuous on (a, b) , f is continuous from the left at a and from the right at b
 D. f is continuous on (a, b) , f is continuous from the left at a and from the left at b

55. If $f'(1) = 4$, $f(1) = 3$ and $g(3) = 5$, then $(g \circ f)'(1)$ is equal to

(1)

A. 5

B. 3

C. 20

D. 15

— END OF EXAMINATION —

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